

# CASPEr@night — Installation and First-Run Guide

(Bruker / TopSpin)

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## 1 What this is

CASPEr@night is a community measurement program: any NMR facility with a Bruker spectrometer can contribute a data point with one 5 mm tube of water and one TopSpin command, `xpy spin_noise_run`. The script records the spin-noise feature of water — the Guéron absorption dip on room-temperature probes, the emission bump on cryoprobes — together with the receiver calibrations needed to interpret it absolutely, and packs everything into a single zip for upload. Across many facilities these bundles map the temperature-contrast law of circuit-spin coupling over the community's probe fleet, and they bank spin-noise-limited noise records for fundamental-sensitivity and dark-matter (CASPEr-style) analyses.

The project page is [night.blanchard-science.com](https://night.blanchard-science.com). The data policy (`DATA_POLICY.md` in the distribution) is short and worth the two minutes — your raw data remain yours, and contributing facilities are co-authors on any network publication their data enter.

### The five-minute version

1. Download the latest release zip: <https://github.com/CASPEr-Night/spin-noise-network/releases/latest>
2. Copy two files onto the spectrometer workstation: `topspin/spin_noise_run.py` → `<TSHOME>/exp/stan/nmr/py/user/` and `topspin/pp/zgnoise2d` → `<TSHOME>/exp/stan/nmr/lists/pp/user/`
3. Fill a 5 mm tube with  $\sim 550 \mu\text{L}$  of plain water (any water — you just have to know what it is), insert it, and open any existing  $^1\text{H}$  dataset in TopSpin.
4. Dry run first, no hardware touched: `xpy spin_noise_run simulate`
5. Real run: `xpy spin_noise_run` — answer the dialogs, **confirm the BSMS field sweep is OFF**, walk away ( $\sim 45$  min default).
6. Upload from any machine with Python 3: `python3 uploader/upload_bundle.py <bundle.zip>` (first time: copy `config.example.json` to `config.json` and paste the endpoint and token the coordinator sent you).

## 2 Requirements

- **A Bruker spectrometer running TopSpin 2.x, 3.x, or 4.x.** (TopSpin 5.0, released 2026, is untested: Bruker documents the Jython layer this software runs in as a standard component alongside the newer CPython interface, so it is expected to work — but we have not verified it.)

If your console runs TopSpin 5, please tell us what happens: you would be our first.) The script is written for TopSpin’s embedded Jython interpreter and avoids anything version-specific — every optional command (`atma`, `topshim`, `pulseval`, `rga`) degrades to an operator dialog when missing, so old consoles work too.

- **Any  $^1\text{H}$ -capable probe** — room-temperature, nitrogen-cooled, or helium cryoprobe. The run asks which and records it.
- **One 5 mm tube of plain water**,  $\sim 550\ \mu\text{L}$ . Tap, distilled, or  $\text{D}_2\text{O}$ -doped all work — *as long as you know the composition*. The  $\text{H}_2\text{O}$ -fraction question in the dialogs is the measurement itself (the spin-noise feature scales with proton density), so if the tube contains  $\text{D}_2\text{O}$ , know the percentage.
- **Python 3.6 or newer on any machine that can reach the internet**, for the upload step only. The spectrometer host itself needs no internet access and no Python packages — carry the zip on a USB stick if needed. Check with:

```
python3 --version
```

- **NTP time synchronization enabled on the TopSpin workstation** (it usually is by default). The run measures your console’s master-clock offset for free by comparing the NTP-disciplined workstation clock against the console-timed acquisition durations — the “clock audit” in the bundle. Nothing on the spectrometer is touched or reconfigured — if your workstation is air-gapped, just mention that when you upload.
- **Disk and time:** the default run writes roughly 200 MB ( $\geq 10$  GB free is comfortable headroom for reruns and longer noise blocks) and takes  $\sim 45$  minutes of magnet time, of which you are needed for the first  $\sim 5$  (the dialogs).

### 3 Download

Get the latest release zip from the GitHub releases page:

```
https://github.com/CASPER-Night/spin-noise-network/releases/latest
```

The zip is the complete `spin-noise-network` distribution: the TopSpin acquisition kit, the uploader, the metadata schema, the protocol and policy documents, and testing tools. Only two files go onto the spectrometer workstation (next section):

- `topspin/spin_noise_run.py` — the acquisition script (Jython, runs inside TopSpin).
- `topspin/pp/zgnoise2d` — the no-pulse pseudo-2D pulse program used for the noise blocks.

Keep the rest of the unpacked folder around on whatever machine will do the upload: the uploader, `uploader/upload_bundle.py`, expects `schema/meta.schema.json` one level up from itself, as in the release layout. If you move the uploader out on its own, pass `--schema <path>` explicitly — otherwise its `--selftest` skips schema validation with a WARN yet still prints `RESULT: PASS`.

### 4 Install (two file copies)

1. Copy `spin_noise_run.py` to `<TSHOME>/exp/stan/nmr/py/user/`. Alternative: in TopSpin type `edpy`, use `File`  $\rightarrow$  `Import...` and pick the file — that lands it in the same place.
2. Copy `pp/zgnoise2d` to `<TSHOME>/exp/stan/nmr/lists/pp/user/`.
3. There is no step 3. No Python packages, no licenses, no network access is needed on the

spectrometer.

<TSHOME> is your TopSpin installation directory — `/opt/topspin4.1.4` or `/opt/topspin3.6.5` on Linux, `C:\Bruker\TopSpin4.1.4` on Windows, etc. If you are unsure, type `set` inside TopSpin or look at the title bar of the TopSpin window.

The script also installs `zgnoise2d` automatically on first run if it can find your TopSpin directory, so step 2 is a belt-and-braces copy — do it anyway if you can.

**Verify TopSpin sees the script:** type `edpy` and `spin_noise_run` should appear in the *user* source directory listing. The pulse program is checked again at run time: if `zg` cannot find it, the script tells you and detects the file once you copy it in by hand (Section 8).

## 5 Verify without touching hardware

Two test modes let you exercise the whole flow before the magnet is involved. Both are started from the TopSpin command line with any  $^1\text{H}$  dataset open (a PROTON demo set is fine — it is only a parameter template).

### 5.1 SIMULATE — the pure dry run

```
xpy spin_noise_run simulate
```

SIMULATE skips all spectrometer commands (and the setup blocks that lead to them) while still running every dialog, the dataset bookkeeping, the `meta.json` writing, and the zip bundling. Answer the dialogs with test values (suggested: institution `Desk Test Lab`, slug `desktest-lab`, sample `distilled water`,  $\text{H}_2\text{O}$  100,  $\text{D}_2\text{O}$  0, duration 30 min, lock OFF, sweep confirmed OFF). Success looks like:

- the greeting dialog shows `*** SIMULATE MODE ... ***`.
- all operator dialogs appear, in the order listed in Section 6.
- the terminal shows `SIMULATE -> ... / SIMULATE: zg mocked ...` lines and no errors.
- the final dialog reports the bundle path, and the zip exists there.

### 5.2 DESKTEST — the stronger check

```
xpy spin_noise_run desktest
```

DESKTEST drives the *real* TopSpin API — the full dialog chain, `GETPAR/PUTPAR` on live parameter sets, `WR/RE` dataset creation, `meta.json` writing, zip bundling — and mocks only the five hardware commands (`atma`, `topshim`, `pulsecal`, `zg`, `rga`). It runs fine on a processing-only TopSpin install (Bruker's free academic license), so you can do it at your desk. Pass criteria, from the release's desk-test checklist:

- terminal shows `DESKTEST -> mocked 'atma', 'topshim', 'pulsecal', 'rga'` — and no manual-fallback dialog appears for any of them, and no Jython traceback anywhere.
- a dataset `SPINNOISE_<yyyymmdd>_<hhmm>` exists with expnos **1, 10, 14, 15, 16, 11, 12, 13**.
- `meta.json` is written twice (dataset directory and bundle) and contains a `software` block with `"script_version": "0.3.0"` (this release), `"schema_version": "1.2"` (the orchestrator's Bruker-only metadata vintage, accepted under the repository's vendor-neutral v2.0 schema), a `script_sha256` fingerprint, and `"run_mode": "desktest"` — plus a `clock_audit` block with one timing entry per acquisition.

- the bundle zip, `SPINNOISE_<date>_<time>/spinnoise_<slug>_<YYYYMMDD_HHMMSS>Z_<hex>.zip`, opens in any zip tool, and the validator passes on any Python-3 machine:

```
python3 uploader/upload_bundle.py <bundle.zip> --selftest
-> RESULT: PASS
```

**Never upload a SIMULATE or DESKTEST bundle** — it contains copies of your demo dataset, not noise data. `meta.json` flags it, but do not rely on the flag — just do not send it.

## 6 The real run

One honest note before the first live run: as of v0.3.0 the script has been desk-tested end to end (a real Jython interpreter against a stubbed TopSpin API, plus the SIMULATE/DESKTEST modes above), but it has not yet run on real spectrometer hardware. We therefore treat the first live run at each facility as a supervised pilot — write to [jwbquantum@gmail.com](mailto:jwbquantum@gmail.com) and we will screen-share through it.

### 6.1 Before you type the command

Insert the water tube (sample changer or manual — the script never ejects or injects), set your usual VT setpoint, and let the sample equilibrate for a few minutes. Tune/match and shimming happen *inside* the run: the script calls `atma` and `topshim` where your system has them and walks you through the manual equivalents (`wobb`, your usual shim routine) where it does not. Open any existing  $^1\text{H}$  dataset, then:

```
xpy spin_noise_run
```

### 6.2 The dialogs, in order

1. **Welcome** — summarizes what the run will do (Start/Cancel).
2. **Your facility (1/5)** — institution, city, country, contact e-mail, so the bundle is attributed correctly.
3. **Facility slug (2/5)** — short machine-readable ID (auto-suggested), used in the bundle file name.
4. **Contact consent** — whether the maintainers may contact you at that address. The e-mail is stored only if you answer yes.
5. **The sample (3/5)** — description,  $\text{H}_2\text{O}$  fraction (%),  $\text{D}_2\text{O}$  fraction (%), additives, tube OD, volume. Answer exactly as the tube is. Tap water with a described history beats an undocumented perfect sample.
6. **Temperature (4/5)** — VT setpoint in K (enter 298 if VT is off and the bore is at room temperature).
7. **Noise-block duration (5/5)** — 30 min (default), 60 min, 180 min, or overnight (10 h). Setup and references add  $\sim 15$  min on top. Longer runs average down the spectrum, and an overnight run on an idle weekend magnet is gold.
8. **Lock state** — turn the lock OFF for the noise block if you can (`lock off` / BSMS LOCK key): lock rf leaks into the  $^1\text{H}$  channel on some systems. Either answer is accepted and recorded.

9. **BSMS field sweep — the one that matters.** Open `bsmsdisp` and *physically verify* SWEEP is OFF before answering. A sweeping field smears the spin-noise feature over several kHz and leaves a perfectly healthy-looking flat noise floor — this silently contaminated an archival dataset in 2022, and the script cannot verify the sweep state on all console generations, so the confirmation is on you. If you cannot confirm, a follow-up dialog offers proceed-flagged or abort. Prefer fixing it to proceeding flagged.
10. **Hardware check** — TopSpin version, console, probe string (pre-filled where the script can detect them) — correct as needed.
11. **Probe type** — room-temperature, nitrogen-cooled, helium cryoprobe, or unknown (this sets which side of the temperature-contrast law your point lands on).
12. **Probe temperatures (optional)** — coil and preamp temperatures in K if you know them (`edte` / cryopanel). Blank is fine.
13. **90-degree pulse** — confirm P90 ( $\mu\text{s}$ ) and its power (dB), pre-filled from `pulsecal` or the current P1. Sanity-check it ( $\sim 7\text{--}15\ \mu\text{s}$  is typical) before confirming.
14. *Noise block auto-start* — after the references the script prints the row count, per-row time, and RG to the status line and starts the pulse-free block by itself 30 s later. There is no dialog to click: *the P90 confirmation above is the walk-away point* (the RG ladder and opening reference run unattended in between,  $\sim 15$  min).
15. **Final notes** — anything unusual during the run (construction, elevator, He fill, lock left on, ...). Blank if all was quiet.
16. **Done** — the bundle path and the one-line upload command.

Two conditional extras: if the script cannot find your TopSpin directory it asks once for the user pulse-program path, and on systems missing `atma/topshim/pulsecal/rga` it asks you to perform that one step manually and press OK.

### 6.3 What it does, and what it will not touch

The run creates one dataset, `SPINNOISE_<date>_<time>`, in your current data directory:

| expno          | role                                                                                    |
|----------------|-----------------------------------------------------------------------------------------|
| 1              | setup: tune/match, shim, P90 calibration                                                |
| 10, 14, 15, 16 | RG ladder: quick $1^\circ$ 1Ds at RG = 1, 8, 64, max                                    |
| 11             | reference (open): $1^\circ$ pseudo-2D, 8 rows $\times$ $\sim 19$ s                      |
| 12             | <b>noise:</b> <code>zgnoise2d</code> , <i>no pulse at all</i> , NS=1/row, RG max stable |
| 13             | reference (close): same as 11 (drift check)                                             |

Pulsing is limited to the standard pulse calibration and  $\sim 1^\circ$  tips at the calibrated observe power — no long or repeated high-power irradiation. The script never changes instrument configuration, never touches your lock or sweep settings silently (it asks you and records your answer), and reads and writes nothing outside the `SPINNOISE_<date>_<time>` dataset plus the two installed files. Complete removal, if you ever want everything gone:

```
rm -rf <DATADIR>/SPINNOISE_<yyyymmdd>_<hhmm>
rm <TSHOME>/exp/stan/nmr/py/user/spin_noise_run.py
rm <TSHOME>/exp/stan/nmr/lists/pp/user/zgnoise2d
```

(Windows: delete the same three paths in Explorer. `<DATADIR>` is the data directory of the template dataset you had open.)

The final dialog prints the full path of the finished bundle, which lands inside the `SPINNOISE_<date>_<time>` dataset directory as `spinnoise_<slug>_<timestamp>Z_<hex>.zip`. You keep this local copy no matter what happens during upload.

## 7 Upload

### 7.1 One-time setup

On whichever machine will do the upload (Python  $\geq$  3.6, internet), go to the `uploader/` folder, copy `config.example.json` to `config.json`, and fill in the values **the coordinator sent your facility privately**:

```
{
  "endpoint_url": "https://...workers.dev/ingest",
  "token":        "<from the coordinator>",
  "facility_slug": "your-facility-slug"
}
```

The endpoint and token arrive by a private channel, never in e-mail chains — treat the token like a password. `config.json` is read locally and never uploaded.

### 7.2 The upload

```
python3 uploader/upload_bundle.py /path/to/spinnoise_*.zip
```

Size never matters: bundles up to 50 MiB go in one request, and anything larger (overnight runs, up to 5 GiB) automatically switches to a chunked upload in 50 MiB parts. Progress is checkpointed next to the zip (`<bundle>.upload-state.json`), so if the network drops or the machine reboots, **rerun the same command and it resumes where it left off**. The uploader never deletes your zip.

Success ends with a line like:

```
RECEIPT: <id>
```

That is the server confirming your bundle arrived intact (its SHA-256 was verified on arrival) and was recorded. Note the id — it is the reference for any follow-up about this bundle.

### 7.3 First test run, or no endpoint yet

If you have no `config.json` (or no internet path at all), the script prints exactly what to do: the zip stays where it is, and you can e-mail or file-transfer it to `jwbquantum@gmail.com` (a share link is fine if it exceeds mail limits). For a first test you can also validate locally without any network:

```
python3 uploader/upload_bundle.py <bundle.zip> --selftest
```

## 8 Troubleshooting

- “No dataset is open” — open any <sup>1</sup>H dataset first (it is only the parameter template).
- **Pulse program not found at zg** — copy `pp/zgnoise2d` into the user pulse-program directory (`<TSHOME>/exp/stan/nmr/lists/pp/user/`) by hand and rerun — the script will detect it.

- **A parmode dialog appears** — some TopSpin versions ask before converting a dataset to 2D. Answer yes/OK — the dataset is fresh, there is nothing to lose.
- **Old TopSpin (2.x)** — everything is written for Jython 2.2-level syntax. If `Avance.incl` is missing, delete the `#include` line in `zgnoise2d` (it is not used).
- **No ATM probe / no rga** — expected and handled: the script asks you to tune/match with wobb (or set RG) manually and confirm in a dialog.
- **A hardware command hangs with no progress for over 5 minutes** — kill the script window. Any acquisition already started finishes on its own and all data stays in `SPINNOISE_<date>_<time>`.
- **A Jython error dialog appears** — the run stopped, but started acquisitions finish and the data stays in `SPINNOISE_<date>_<time>`. Copy the error text verbatim into a support issue (below) — a stuck run reported well is worth almost as much as a clean one.
- **Facility Python 3 older than 3.6** — run the selftest and upload from any other machine (copy the zip on a stick). The uploader is stdlib-only by design, so please do not pip-install anything on the spectrometer workstation.

**Where to get help:** open a facility-support issue at <https://github.com/CASPER-Night/spin-noise-network/issues/new?template=facility-support.yml> (“don’t know” is always an acceptable answer on that form), or write to [jwbquantum@gmail.com](mailto:jwbquantum@gmail.com).

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Data policy: your raw data remain yours, and contributing facilities are co-authors on network publications their data enter — full terms in `DATA_POLICY.md` and at [night.blanchard-science.com](http://night.blanchard-science.com).